

Carnegie Mellon University

**FPN-Mamba: A Feature Pyramid Network with
Bidirectional State Space Models for
Medulloblastoma Classification**

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INTRODUCTION

The Clinical Problem

20%

of pediatric brain tumors

*A missed case is not a metric error.
It is a child without a diagnosis.*

- Most common malignant brain tumor in children
- Missed diagnosis = delayed treatment, worse outcomes
- Hard to detect: small lesions, low contrast, similar appearance to other tumors
- MRI is the primary tool, but interpretation is difficult even for experts



RELATED WORKS

Method	Multi-scale	Global context	Complexity
CNN + attention Rasheed 2024, Apostolopoulos 2023	No	No	$O(L^2)$
FPN Lin et al. 2017	Yes	No	$O(L)$
Mamba SSM Gu & Dao 2023	No	Yes	$O(L)$
InceptionNet Fang et al. 2025, our baseline	Partial	Single scale	$O(L^2)$
FPN-Mamba (ours) classification	Yes	Yes	$O(L)$

Gap: no work combines both for MB classification

RELATED WORKS

Why Improve a Strong Baseline?

InceptionNet

Fang et al. 2025

98.07%

Accuracy

99.41%

AUC

96.03%

Recall

Single-scale limitation

Attention applied to one coarse feature map small, subtle lesions are structurally invisible

False negatives cluster here

Fang et al. admit FNs concentrate on small lesions, low contrast, atypical location

Goal: clinical, not benchmarking

Reducing FN rate from ~500 to 300 means fewer children misdiagnosed

METHODOLOGY

Dataset and Setup

761

total images

131 vs 630

MB vs Non-MB

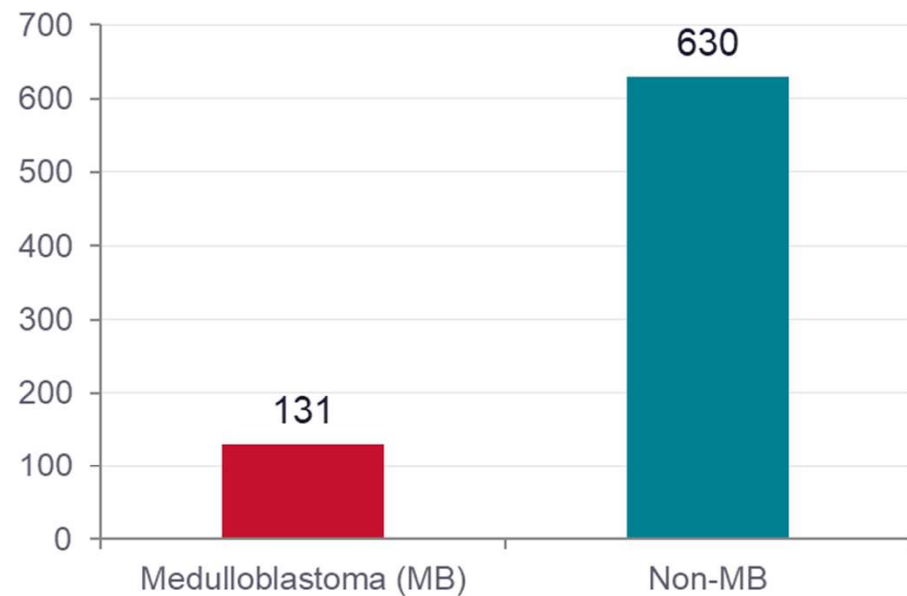
5-fold

stratified CV

4.85×

pos_weight correction

Class Distribution



Heavier augmentation applied to MB class only to address imbalance

METHODOLOGY

Architecture: FPN-Mamba



Multi-scale

Each pyramid level sees the tumor at a different resolution — small lesions are not lost

LocalityMixing

DWConv for local detail + Bidirectional Mamba for global context, blended by a learned gate

$O(L)$ complexity

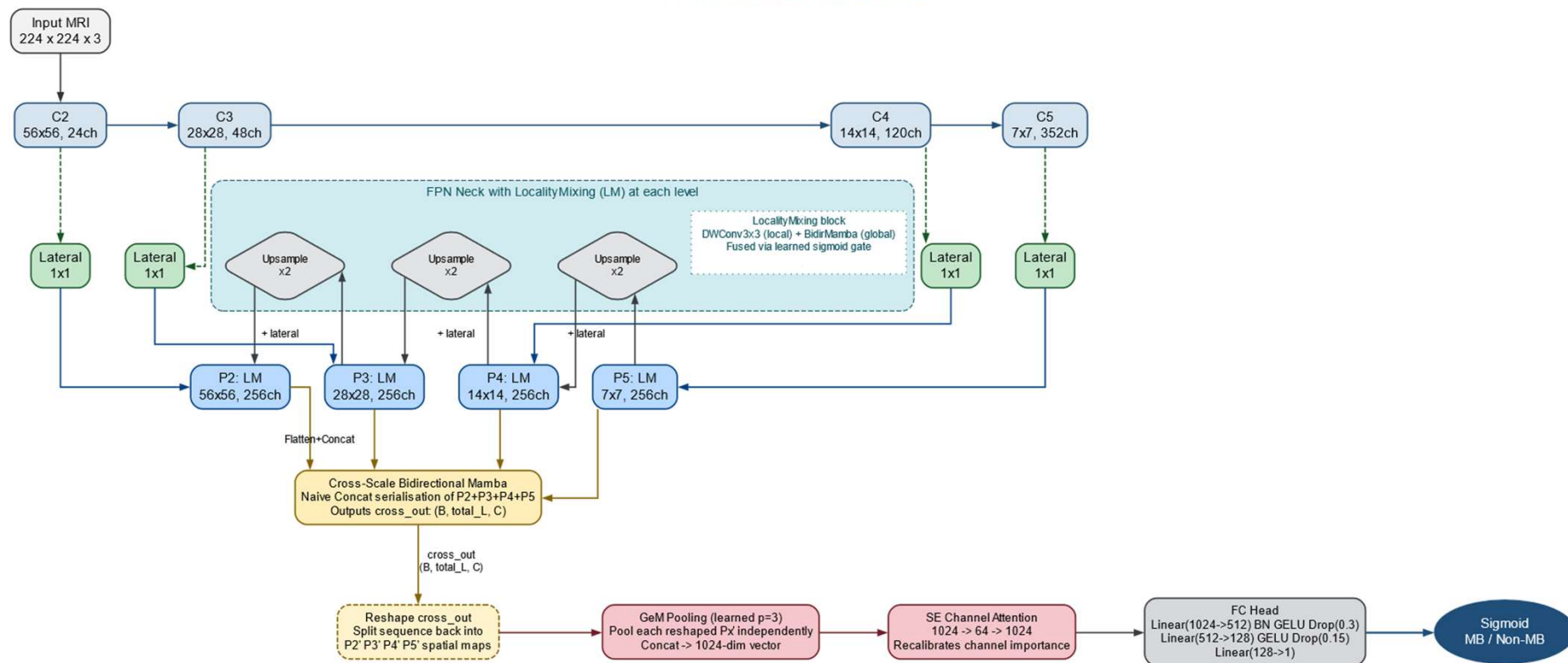
Mamba scales linearly — global context is feasible at fine-resolution levels unlike $O(L^2)$ attention

GeM + SE

Emphasises high-activation regions; SE recalibrates which channels matter most before classification

METHODOLOGY

FPN-Mamba Architecture



RESULTS

Quantitative Comparison

4 of 5

metrics beat baseline

+2.80pp

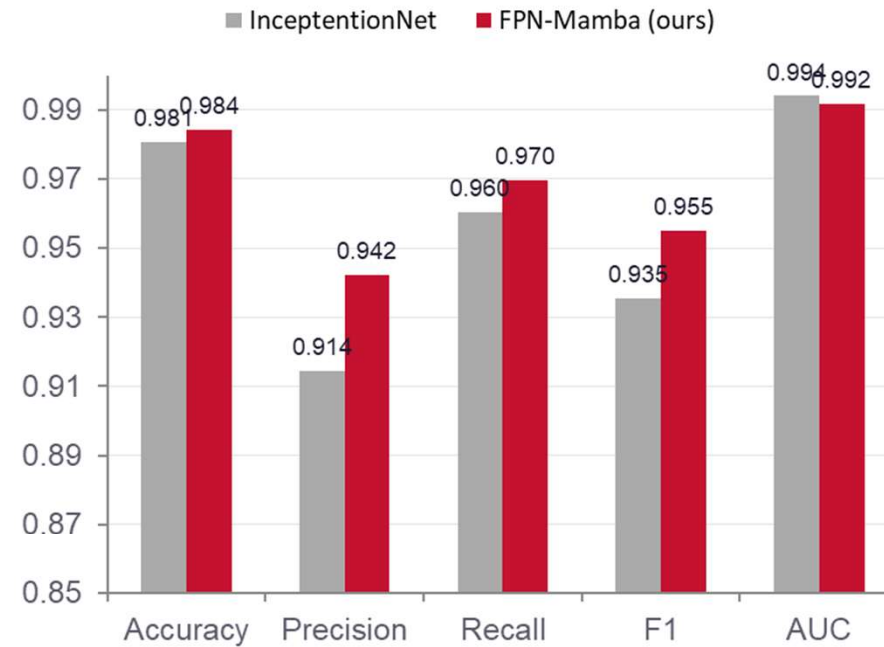
precision improvement

+1.97pp

F1 improvement

3 FN

false negatives (vs ~5)



CONCLUSION

Summary and Future Work

Multi-scale detection

FPN ensures tumors at every size are represented. Nothing is discarded before the classifier sees it.

Linear-complexity context

Mamba provides global context at fine-resolution levels — where small tumors live — without quadratic cost.

Clinical objective

3 false negatives vs ~5 baseline. Metrics are a byproduct. Fewer missed diagnoses is the goal.

Limitations: No public external MB dataset · No formal ablation study completed

Next: Grad-CAM at each pyramid level · Formal ablation study · Larger institutional dataset

REFLECTION

Concepts Learnt

Feature Pyramids in practice

The FPN concept taught in class directly shaped the architecture, seeing it applied to real medical imaging showed how powerful multi-scale thinking is beyond object detection.

Research requires depth

Extensive literature review led to better design choices. Understanding the baseline's limitations was as important as building our own solution.

Beyond the benchmark

Research in healthcare is not about improving a score. Every false negative has a human cost. That reframing changed how we thought about every design decision.



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